

SHETH L.U.J & SIR M.V COLLEGE OF SCIENCE
SUBJECT : Cyber & Information Security

AIM : Implement algorithms to generate and verify message authentication codes (MACs) for ensuring data integrity and authenticity.

CODE:

```
import hmac
import hashlib

secret_key = input("Enter Secret Key: ").encode()

message = input("Enter Message: ").encode()

mac = hmac.new(secret_key, message, hashlib.sha256).hexdigest()

print("\nGenerated MAC:")
print(mac)

print("\n===== Verification =====")

received_message = input("Enter Received Message: ").encode()

received_mac = input("Enter Received MAC: ")

new_mac = hmac.new(secret_key, received_message,
hashlib.sha256).hexdigest()

if hmac.compare_digest(new_mac, received_mac):
    print("\nMAC Verified Successfully")
    print("Message is Authentic")
    print("Data Integrity Maintained")
else:
    print("\nVerification Failed")
    print("Message has been Tampered")
```

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OUTPUT

```
File Edit Selection View Go Run Terminal Help cybersecurity Sign In
EXPLORER T108-PRAC3.py x
CYBERSECURITY
rsa_cli.py
rsa_gui.py
T108-caesar-cli.py
T108-CAESAR-RAILFE...
T108-CYBER & INFOR...
T108-CYBER PRAC2.pdf
T108-PRAC3.py
T108-railfence-cli.py
T108-PRAC3.py x
1 import hmac
2 import hashlib
3
4 secret_key = input("Enter Secret Key: ").encode()
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS D:\T108\cybersecurity> & C:\Users\itlab\AppData\Local\Programs\Python\Python314\python.exe d:/T108/cybersecurity/T108-PRAC3.py
Enter Secret Key: batman123
Enter Message: hello batman

Generated MAC:
64fbb0a320a7bb2666c34ddf84a2062e96d0a89955d9b2bfad7b7ad8c18db64b

----- Verification -----
Enter Received Message: bat
Enter Received MAC: 64fbb0a320a7bb2666c34ddf84a2062e96d0a89955d9b2bfad7b7ad8c18db64b

Verification Failed
Message has been Tampered
PS D:\T108\cybersecurity> []

PS D:\T108\cybersecurity> & C:\Users\itlab\AppData\Local\Programs\Python\Python314\python.exe d:/T108/cybersecurity/T108-PRAC3.py
Enter Secret Key: hello123
Enter Message: hello superman

Generated MAC:
f06899980391e73cd8a0ee96e8c0bfeff65d5b3088ec134bde2654e840fcd1

----- Verification -----
Enter Received Message: hello superman
Enter Received MAC: f06899980391e73cd8a0ee96e8c0bfeff65d5b3088ec134bde2654e840fcd1

MAC Verified Successfully
Message is Authentic
Data Integrity Maintained
PS D:\T108\cybersecurity> []
```